Chapter 3, Section 7

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1. Let X be an integral projective scheme of dimension $\geqslant 1$ over a field k, and let $\mathcal{L}$ be an ample invertible sheaf on X. Then $H^0(X, \mathcal{L}^{-1}) = 0$.

Proof. A global section of $\mathcal{L}^{-1}$ defines an $\mathcal{O}_X$-module morphism

$$\varphi : \mathcal{L} \to \mathcal{O}_X,$$

and hence induces morphisms

$$\varphi^{\otimes n} : \mathcal{L}^{\otimes n} \to \mathcal{O}_X$$

for all $n > 0$. Since $\mathcal{L}$ is locally free of rank one, φ is locally given by multiplication by a section of $\mathcal{O}_X$. If for some $x \in X$ the stalk map $\varphi_x^{\otimes n}$ is not injectve, it must be the zero map. As X is integral, this forces φ to vanish at the generic point, hence $\varphi = 0$. Assume therefore $\varphi \neq 0$. Since the global section functor is left exact, $\varphi^{\otimes n}$ induces an injective map

$$\Gamma(\varphi^{\otimes n}) : H^0(X, \mathcal{L}^{\otimes n}) \to H^0(X, \mathcal{O}_X).$$

However, $H^0(X, \mathcal{O}_X)$ is finite dimensional over k (5.2), while for $n \gg 0$, we have

$$\dim_k H^0(X, \mathcal{L}^{\otimes n}) = \chi(\mathcal{L}^{\otimes n}),$$

which is a polynomial in n (5.3), and hence eventually exceeds $\dim_k H^0(X, \mathcal{O}_X)$. This is a contradiction, and therefore $\varphi = 0$. $\square$

2. Let $f : X \to Y$ be a finite morphism of projective schemes of the same dimension over a field k, and let ω_Y° be a dualizing sheaf for Y.

(a) Show that $f^! \omega_Y^\circ$ is a dualizing sheaf for X, where $f^!$ is defined as in (Ex. 6.10).

(b) If X and Y are both nonsingular, and k algebraically closed, conclude that there is a natural trace map $t : f_* \omega_X \to \omega_Y$.

Proof.

(a) We must first define a trace map $t_X : H^n(X, f^! \omega_Y^\circ) \to k$. By (Ex. 4.1), we have a natural isomorphism

$$H^n(X, f^! \omega_Y^\circ) \xrightarrow{\sim} H^n(Y, f_* f^! \omega_Y^\circ).$$

There is a natural morphism of $\mathcal{O}_Y$-modules $f_* f^! \omega_Y \to \omega_Y$, which induces a map between cohomology

$$H^n(Y, f_* f^! \omega_Y^\circ) \to H^n(Y, \omega_Y^\circ).$$

Composing with the trace map of ω_Y° gives us a trace map for $f^! \omega_Y^\circ$. It remains to show t_X satisfies the universal property, so let $\mathcal{F}$ be a coherent sheaf on X. Then $f_* \mathcal{F}$ is also a coherent sheaf on Y, so by (Ex. 6.10), we have

$$\mathrm{Hom}_X(\mathcal{F}, f^! \omega_Y) \cong \mathrm{Hom}_Y(f_* \mathcal{F}, \omega_Y) \cong H^n(Y, f_* \mathcal{F})^\vee \cong H^n(X, \mathcal{F})^\vee.$$

(b) By (Ex. 6.10), defining a morphim $t : f_* \omega_X \to \omega_Y$ is the same as defining a morphism $\omega_X \to f^! \omega_Y$, so we can let t be the morphism corresponding to a fixed (unique) isomorphism $\omega_X \to f^! \omega_Y$.

$\square$

3. Let $X = \mathbb{P}^n_k$. Show that $H^q(X, \Omega^p_X) = 0$ for $p \neq q$, k for $p = q$, $0 \leqslant p, q \leqslant n$.

Proof. Consider the Euler sequence of (II, 8.13)

$$0 \longrightarrow \Omega_X \longrightarrow \mathcal{O}_X(-1)^{\oplus n+1} \longrightarrow \mathcal{O}_X \longrightarrow 0.$$

For each $0 \leqslant p \leqslant n$, let $\mathcal{G}^{p+1} = \bigwedge^{p+1} \mathcal{O}_X(-1)^{\oplus n+1}$, which is isomorphic to the direct sum of $\binom{n+1}{p+1}$ copies of $\mathcal{O}_X(-1)$. By (II, Ex. 5.16c), for each $0 \leqslant p \leqslant n$, there exists a filtration $\{\mathcal{F}^q\}$ of $\mathcal{G}^{p+1}$ such that

$$\mathcal{F}^q/\mathcal{F}^{q+1} \cong \Omega^q_X \otimes \bigwedge^{p+1-q} \mathcal{O}_X = \begin{cases} \Omega^q_X & q = p, p+1 \\ 0 & \text{otherwise.} \end{cases}.$$

Since $\mathcal{O}_X$ has rank one,

$$\mathcal{G}^{p+1} = \mathcal{F}^0 = \mathcal{F}^1 = \cdots = \mathcal{F}^p, \quad \mathcal{F}^p/\mathcal{F}^{p+1} \cong \Omega^p_X, \quad \mathcal{F}^{p+1} \cong \Omega^{p+1}_X,$$

which can be put into the short exact sequence

$$0 \longrightarrow \Omega^{p+1}_X \longrightarrow \mathcal{G}^{p+1} \longrightarrow \Omega^p_X \longrightarrow 0.$$

Taking the long exact sequence of cohomology gives

$$\cdots \longrightarrow H^i(X, \mathcal{G}^{p+1}) \longrightarrow H^i(X, \Omega^p_X) \longrightarrow H^{i+1}(X, \Omega^{p+1}_X) \longrightarrow H^{i+1}(X, \mathcal{G}^{p+1}) \longrightarrow \cdots$$

Direct sums commute with cohomology (2.9), and $H^i(X, \mathcal{O}_X(-1)) = 0$ for all i, which means $H^i(X, \mathcal{G}^{p+1}) = 0$, so from the long exact sequence, we obtain the isomorphisms

$$H^i(X, \Omega^p_X) \cong H^{i+1}(X, \Omega^{p+1}_X).$$

Hence, if $q \geqslant p$,

$$H^q(X, \Omega^p_X) \cong H^{q-p}(X, \mathcal{O}_X) = \begin{cases} k & p = q, \\ 0 & \text{otherwise.} \end{cases}$$

The other Hodge numbers are obtained via Serre duality. $\square$

4. *The Cohomology Class of a Subvariety.* Let X be a nonsingular projective variety of dimension n over an algebraically closed field k. Let Y be a nonsingular subvariety of codimension p (hence dimension $n - p$). From the natural map $\Omega_X \otimes \mathcal{O}_Y \to \Omega_Y$ of (II, 8.12) we deduce a map $\Omega^{n-p}_X \to \Omega^{n-p}_Y$. This induces a map on cohomology $H^{n-p}(X, \Omega^{n-p}_X) \to H^{n-p}(Y, \Omega^{n-p}_Y)$. Now $\Omega^{n-p}_Y = \omega_Y$ is a dualizing sheaf for Y, so we have the trace map $t_Y : H^{n-p}(Y, \Omega^{n-p}_Y) \to k$. Composing, we obtain a linear map $H^{n-p}(X, \Omega^{n-p}_X) \to k$. By (7.13) this corresponds to an element $\eta(Y) \in H^p(X, \Omega^p_X)$, which we call the *cohomology class* of Y.

(a) If $P \in X$ is a closed point, show that $t_X(\eta(P)) = 1$, where $\eta(P) \in H^n(X, \Omega^n)$ and t_X is the trace map.

(b) If $X = \mathbb{P}^n$, identify $H^p(X, \Omega^p)$ with k by (Ex. 7.3), and show that $\eta(Y) = (\deg Y) \cdot 1$, where $\deg Y$ is its *degree* as a projective variety (I, §7).

(c) For any scheme X of finite type over k, we define a homomorphism of sheaves of Abelian groups $d\log : \mathcal{O}^*_X \to \Omega_X$ by $d\log(f) = f^{-1}df$. Here $\mathcal{O}^*_X$ is a group under multiplication, and Ω_X is a group under addition. This induces a map on cohomology $\operatorname{Pic} X = H^1(X, \mathcal{O}^*_X) \to H^1(X, \Omega_X)$ which we denote by c.

(d) Returning to the hypotheses above, suppose $p = 1$. Show that $\eta(Y) = c(\mathcal{L}(Y))$, where $\mathcal{L}(Y)$ is the invertible sheaf corresponding to the divisor Y.